

Monday, Feb 23rd

— Quiz —

Wednesday, Feb 25

Least Squares Problem

(I) Continuous

(II) Discrete

(I) continuous function: given a function $f(x)$ provide a "simple" approx

$$\text{ex } f(x) \approx f(x_0) + f'(x_0)(x - x_0)$$

(II) discrete function: given data $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$

find a "simple" representation for data

$$\text{ex } P_n(x) = \sum_{i=0}^n y_i \prod_{k=0, k \neq i}^n \frac{(x - x_k)}{(x_i - x_k)}$$

note: high degree polynomials can suffer from Runge phenomena

→ why we want to reduce the degree of an approx poly.

(II) discrete case - say we want to use a 1st degree polynomial: $P_1(x) = ax + b$

... to simplify...

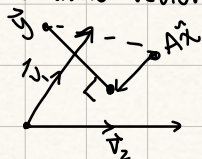
Assume: $(x_0, y_0), (x_1, y_1), \dots, (x_3, y_3)$

Want: $ax_0 + b = y_0$ $ax_1 + b = y_1$ $ax_2 + b = y_2$ $ax_3 + b = y_3$

$$\begin{bmatrix} x_0 & 1 \\ x_1 & 1 \\ x_2 & 1 \\ x_3 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} y_0 \\ y_1 \\ y_2 \\ y_3 \end{bmatrix} \quad \left. \vphantom{\begin{bmatrix} x_0 & 1 \\ x_1 & 1 \\ x_2 & 1 \\ x_3 & 1 \end{bmatrix}} \right\} A\vec{x} = \vec{b} \text{ (linear system)}$$

view $A\vec{x}$ as linear combination of column vectors of A

$$a \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix}_{\vec{v}_1} + b \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}_{\vec{v}_2}$$



solution potentially doesn't exist b/c overdetermined

note: our system may not have a solution

→ how might we approx. the solution

↳ answer: choose $\tilde{x}$ st $A\tilde{x} \perp (\vec{y} - A\tilde{x})$

Review - inner products

for $\vec{x}, \vec{y}$ in $\mathbb{R}^n$, $\langle \vec{x}, \vec{y} \rangle = \vec{x}^T \vec{y}$

note: when $\vec{x} \perp \vec{y}$, $\langle \vec{x}, \vec{y} \rangle = 0$

1) $\langle \vec{x}, \vec{x} \rangle = \|\vec{x}\|^2 \geq 0$

2) $\langle \vec{x}, \vec{x} \rangle = 0$ iff $\vec{x} = 0$

3) $\langle \vec{x} + \vec{z}, \vec{y} \rangle = \langle \vec{x}, \vec{y} \rangle + \langle \vec{z}, \vec{y} \rangle$

∴ want $\langle A\tilde{x}, \vec{y} - A\tilde{x} \rangle = 0$ where $\tilde{x} = \begin{bmatrix} a \\ b \end{bmatrix}$

→ $(A\tilde{x})^T (\vec{y} - A\tilde{x}) = 0$

→ $(\tilde{x}^T A^T) (\vec{y} - A\tilde{x}) = 0$

→ $\langle \tilde{x}, A^T(\vec{y} - A\tilde{x}) \rangle = 0 \rightarrow$ must have $A^T(\vec{y} - A\tilde{x}) = 0 \rightarrow$ normal eqns for the approx to $\vec{y}$

$$A^T A \tilde{x} = A^T \vec{y}$$

note: $A^T A$ is symmetric → if A is $m \times n$, then $A^T A$ is $n \times n$

ex find the best line to approx

$(-1, 1), (0, 0), (1, 0), (2, -2)$

form $A\tilde{x} = \vec{y}$ where

$$A = \begin{bmatrix} -1 & 1 \\ 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{bmatrix} \quad \tilde{x} = \begin{bmatrix} a \\ b \end{bmatrix} \quad \vec{y} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ -2 \end{bmatrix}$$

$$A^T A \tilde{x} = A^T \vec{y}$$

$$\begin{bmatrix} 6 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} -5 \\ -1 \end{bmatrix}$$

$$f(x) = -\frac{9}{10}x + \frac{1}{5}$$

→ another perspective: min. the dist btw $\text{span}[\vec{v}_1, \vec{v}_2]$ & $\vec{y}$

$$\vec{y} - A\vec{x} \rightarrow \sum_{i=1}^m (y_i - A\vec{x}_i) \rightarrow \underbrace{\sum_{i=1}^m (y_i - (ax_i + b))^2}_{\text{minimize}}$$

Friday 2/27

$$\langle A\vec{x}, \vec{y} - A\vec{x} \rangle = 0 \Rightarrow A^T(\vec{y} - A\vec{x}) = 0$$

ex: $\left. \begin{array}{l} a_1 x_1 + a_0 = b_1 \\ a_1 x_2 + a_0 = b_2 \\ \vdots \\ a_1 x_m + a_0 = b_m \end{array} \right\} \begin{array}{l} \text{y values} \\ Q(a_0, a_1) = \sum_{i=1}^m (b_i - (a_1 x_i + a_0))^2 \\ A = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix}, \vec{b} = \vec{y} \end{array}$

$$\rightarrow \frac{\partial Q}{\partial a_0} = 0, \frac{\partial Q}{\partial a_1} = 0$$

$$\rightarrow 2 \sum_{i=1}^m (b_i - (a_1 x_i + a_0))(-1) = 0$$

$$2 \sum_{i=1}^m (b_i - (a_1 x_i + a_0))(-x_i) = 0$$

Rearrange to get b_i on other side:

$$\begin{aligned} a_1 \sum_{i=1}^m x_i + a_0 m &= \sum_{i=1}^m b_i \\ a_1 \sum_{i=1}^m x_i^2 + a_0 \sum_{i=1}^m x_i &= \sum_{i=1}^m x_i b_i \end{aligned} \quad \left. \begin{array}{l} \\ \end{array} \right\} \begin{array}{l} \text{show if you} \\ \text{take } A^T \text{ it's equiv.} \end{array}$$

$$A^T A \vec{x} = A^T \vec{b}$$

② What if data isn't well described by linear model?

Approx. data (x_i, y_i) $i=1, 2, 3, \dots, m$ w/ polynomial

$$P_n(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

with $n+1 < m$ (overdetermined system).

→ goal: minimize

$$\sum_{i=1}^m (y_i - P_n(x_i))^2 = \langle \vec{b} - A\vec{x}, \vec{b} - A\vec{x} \rangle$$

$$\begin{array}{c} m \\ \left\{ \begin{array}{c} \overbrace{\begin{bmatrix} 1 & x_1 & x_1^2 & \dots & x_1^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_m & x_m^2 & \dots & x_m^n \end{bmatrix}}^{n+1} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix} \\ \underbrace{\begin{bmatrix} 1 & x_1 & x_1^2 & \dots & x_1^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_m & x_m^2 & \dots & x_m^n \end{bmatrix}}_A \underbrace{\begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{bmatrix}}_{\vec{x}} = \underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}}_{\vec{b}} \end{array} \right. \\ \text{Vandermonde Matrix} \end{array}$$

ex) find best parabola to approx the data

$$(-1, 1), (0, 0), (1, 0), (2, -2)$$

$$p_2(x) = a_2 x^2 + a_1 x + a_0$$

$$p_2(x_0) = a_2 - a_1 + a_0 = 1$$

$$p_2(x_1) = a_0 = 0$$

$$p_2(x_2) = a_1 + a_0 = 0$$

$$p_2(x_3) = 4a_2 + 2a_1 + a_0 = -2$$

$$= \begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \\ 4 & 2 & 1 \end{bmatrix} \begin{bmatrix} a_2 \\ a_1 \\ a_0 \end{bmatrix} = \begin{bmatrix} -7 \\ 5 \\ -1 \end{bmatrix}$$

Solve using mathematica

computing the residual: $\|\vec{r}\|_2 = \|\vec{b} - A\vec{x}\|_2$

ill-conditioning of $A^T A$

Solving $A^T A \vec{x} = A^T \vec{b}$ is **not practical** b/c $\text{cond}(A^T A)$ large

QR factorization solves normal eqn w/o forming $A^T A$

(ex $\sum_{i=0}^7 a_i x^i \dots \text{cond}(A^T A) = 10^{19}$)